

ASSIGNMENT BRIEF

Module title	Artificial Intelligence
Module code	COM 757
Module leader	
Assessment title	Coursework Task 1
Launch date	
Submission deadline	2026-01-09
Expected date for return of marks and feedback	<i>Within 15 working days</i>
Module outcomes assessed	1,2
Assessment weighting	50%
Word count (if relevant)	2000 words (+/-10%)
Assessment task details - provide a description of the task	
Task Description Students are required to select a real-world problem domain where Artificial Intelligence (AI) could deliver meaningful impact. Examples may include (but are not limited to): fraud detection, personalised healthcare, misinformation detection, autonomous navigation, sustainable agriculture. Students are encouraged to select a domain they find professionally relevant or intellectually engaging, provided it allows sufficient scope for critical analysis. The report must include the following components: Problem Context (20%) - Clearly define the chosen domain and describe the specific problem within that domain. - Justify why the problem is important in practice, supported with evidence such as statistics, industry reports, or case examples. - Show awareness of both the opportunities and the risks of introducing AI into this domain. Critical Review of Research (20%)	

- Conduct a critical survey of at least 10 high-quality academic sources (IEEE, ACM, Springer, or equivalent).
- Compare and contrast how different AI approaches have been applied to address similar or related challenges.
- Highlight their strengths, weaknesses, and underlying assumptions.
- Identify clear gaps in existing research and explain how these gaps open opportunities for innovation or improved solutions.
- Avoid purely descriptive writing: your review should demonstrate evaluation, synthesis, and critique.

Conceptual AI Solution (40%)

- Propose a written conceptual system design that could address the defined problem.
- The design must be supported by a clear conceptual diagram or architecture illustrating the key components.
- Include details such as:
 - Data inputs and sources (structured, unstructured, multimodal, etc.)
 - Knowledge representation (e.g., rules, models, hybrid approaches)
 - Decision-making logic (e.g., inference mechanisms, learning algorithms, reasoning approaches)
 - Evaluation metrics for success (accuracy, fairness, transparency, efficiency, etc.)
- The focus should be on technical coherence and originality — no coding or software implementation is required.

Ethical & Societal Reflection (10%)

- Provide a critical discussion of the ethical, legal, and societal risks associated with your proposed solution.
- Consider issues such as bias, explainability, privacy, accountability, inclusivity, and long-term societal impact.
- Suggest mitigation strategies and best practices to ensure that your AI system remains trustworthy and responsibly deployed.
- This section should not be generic — ensure that the reflections are grounded in the chosen domain and your proposed design.

Presentation & Communication (10%)

- The report should be well-structured, coherent, and professional.

- Use clear headings, logical progression of arguments, and integrate conceptual diagrams effectively.
- Follow academic conventions for citation and referencing consistently (IEEE or Harvard).
- Write for an informed but non-expert academic audience, balancing technical detail with clarity.

Submission instructions - What should be the format of the submission? / Where should it be submitted?

Submit a single Word or PDF document online via the Turnitin link in Moodle.

The report must use IEEE referencing consistently.

Diagrams should be embedded within the text (not as appendices).

Hints and tips

Start with a clear domain choice early.

Focus on critical comparison of methods, not just description.

Ensure your conceptual design is technically coherent (not generic).

Discuss ethical implications in a balanced and applied way.

Marking and moderation

moderation and second marking will be done according to the university policy.

Employability Skills Applied

On successful completion of this module, a student will have had opportunities to demonstrate achievement of the following Employability Skills;

Tick all that apply.

CORE ATTRIBUTES

Engaged	X
Creative	X
Enterprising	X
Ethical	X

KEY ATTITUDES

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Wrexham University

Commitment	X
Curiosity	X
Resilient	
Confidence	
Adaptability	X
PRACTICAL SKILLSETS	
Digital fluency	X
Organisation	
Leadership and team working	
Critical thinking	X
Emotional intelligence	
Communication	X

Marking criteria
Marking Rubric is separately available on Moodle.

ASSIGNMENT BRIEF

Module title	Artificial Intelligence
Module code	COM 757
Module leader	
Assessment title	Coursework Task 2
Launch date	
Submission deadline	2026-01-15
Expected date for return of marks and feedback	<i>Within 15 working days</i>
Module outcomes assessed	2,3
Assessment weighting	50%
Word count (if relevant)	2000 words (+/-10%)
Assessment task details - provide a description of the task	
Task Description Students are required to conduct a critical evaluation of past case studies where AI applications failed or caused significant controversy. These failures may relate to technical limitations, ethical concerns, legal disputes, or societal rejection. The report must include: Case Study Selection & Description (20%) Choose three real-world AI case studies where the deployment or adoption of AI failed. Examples might include: COMPAS recidivism prediction bias in the US justice system Microsoft's "Tay" chatbot disaster Self-driving car accidents (Uber, Tesla)	

Provide context (timeline, domain, goals, stakeholders).

Critical Evaluation of Methods (30%)

Analyse what AI methods/approaches were used in each case.

Critically evaluate why they failed (e.g., poor data quality, lack of explainability, overfitting, bias, weak governance).

Compare the case studies to contemporary research — how are methods being improved today to avoid similar failures?

STEEPLED Analysis (20%)

Discuss the wider societal, technological, economic, environmental, political, legal, ethical, and demographic consequences of the failures.

Show how AI missteps impacted trust, regulation, and adoption.

Future Lessons & Research Implications (20%)

Reflect on what emerging AI methods (e.g., explainable AI, fairness-aware ML, federated learning) could have prevented or mitigated these failures.

Provide evidence-based insights into how future research is addressing these limitations.

Presentation & Communication (10%)

The report should be well-structured, coherent, and professional.

Use clear headings, logical progression of arguments, and integrate conceptual diagrams effectively.

Follow academic conventions for citation and referencing consistently (IEEE or Harvard).

Write for an informed but non-expert academic audience, balancing technical detail with clarity.

Submission instructions - What should be the format of the submission? / Where should it be submitted?

Submit a single Word or PDF document online via the Turnitin link in Moodle.

The report must use IEEE referencing consistently.

Diagrams should be embedded within the text (not as appendices).

Hints and tips

Avoid pure storytelling — be critical and analytical.

Link past failures to contemporary/emerging methods

Use STEEPLED to frame societal consequences.

Be evidence-driven with citations from both academic sources and credible reports.

Marking and moderation

moderation and second marking will be done according to the university policy.

Employability Skills Applied

On successful completion of this module, a student will have had opportunities to demonstrate achievement of the following Employability Skills;

Tick all that apply.

CORE ATTRIBUTES

Engaged	X
Creative	X
Enterprising	X
Ethical	X

KEY ATTITUDES

Commitment	X
Curiosity	X
Resilient	
Confidence	
Adaptability	X

PRACTICAL SKILLSETS

Digital fluency	X
Organisation	
Leadership and team working	
Critical thinking	X
Emotional intelligence	
Communication	X

Marking criteria
Marking Rubric is separately available on Moodle.